# Assignment TACD/ATDS 2025/2026: first leg

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## Warning

This is the first leg of the assignment. A second leg including NLP should also be out soon.

## Objectives

Recommender systems are an important application of data science and machine learning. It is unthinkable for a nowadays serious web based business not to have a data-driven recommender system approach.

In the first leg of this assignment you will take the movielens small data set and try the different data-driven approaches to RS that we have seen.

## Dataset

The data is the same suggested in one of the labs, the ml-latest-small. You can find it here.

## Tasks

This data science problem should be approached by following the CRISP-DM methodology (http://jbusse.de/2019_ws_dsci/crisp-dm_phases-tasks-outputs.html).

1. Business Understanding
   a. Describe the business problem, propose success criteria and see how it can be translated into machine learning problems.
2. Data Understanding
   a. Describe the characteristics of the data, focusing on the table `ratings`. Perform the required explorations and visualizations.Characterize the number of items and users. Check and visualize the distribution of basket size. Does it follow a power law?
3. Data Transformation
   a. Which transformations do you need for your data in order to apply the machine learning you defined above?
4. Modeling
   a. Do some probing exercises: Find a small enough subsample with 20% of users to facilitate testing approaches before you deep dive into the full data. Check that the sample looks representative in terms of data distribution. Separate a validation set for this one.
      - Try the approaches we have studied (AR, CF, MFPQ, BPR, NCF) just to make thinks work. Measure Precision and Recall at 3 for each of them, when possible, for a binary scenario and for a non-binary scenario.
      - Fine tune the hyperparameters some of them have (number of neighbours, factors, etc.)
      - Visualize how results change with different factors. Before you do any visualization you should always have a question you want to answer with the visualization.

- Add regularization ot the loss functions of the aprroaches that have loss functions. See if regularization changes, fixing the other hyperparameters, with different values of $\lambda$
- What would you propose as best solution under these results?

b. Work with the whole data: Separate training, evaluation and test set for the whole data.
- Perform a similar evaluation but focusing on the best results of the previous round.
- Try 3 arbitrary configurations and compare them with yours
- Show Precision-Recall curves for the 3 or 4 best approaches.

c. Add content data (to be specified later)

During the whole process take notes, always identify the questions you want to answer and think before you act: "why is this plot or this transformation useful". You can perform some operations just for the sake of training but you should be aware of that.

The result is a **report** in the form of a **notebook** with clear explanatory text and code that works showing results. The report should be clear, as concise as possible and it should be easy to read and to follow. You will be telling the story of your approach to this problem, so it should have a good narrative flow. Always explain what you are doing, why you are doing it, what are the results and what do you take from those results.

Always think of the business problem and of future operational conditions that may limit applicability of the model. Not everything that we can do with a dataset is doable under operational conditions.

Prepare a succint 7 slide presentation with the main lessons and results.

## 2nd leg

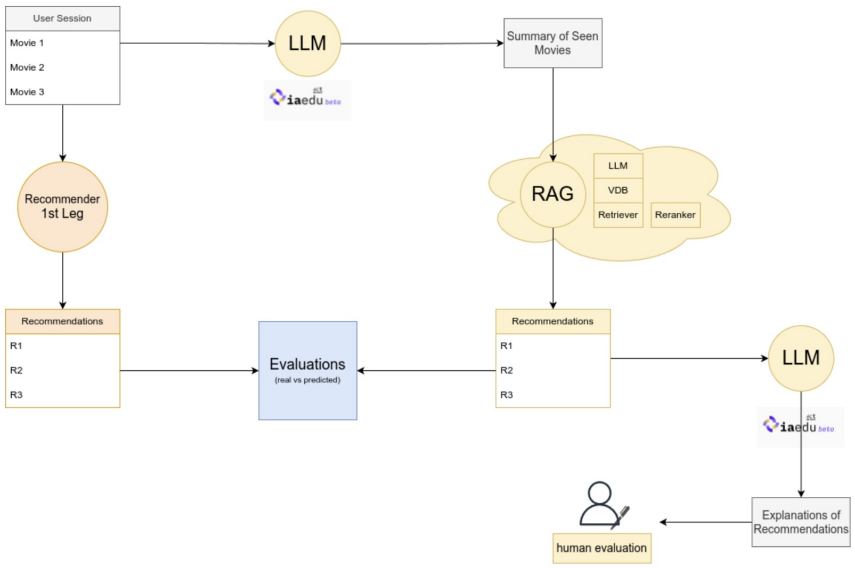



Figure 1: Conceptual diagram of the full assignment

The second leg of the work extends the recommendation systems developed in the first leg by introducing Retrieval-Augmented Generation (RAG) and information retrieval methods. While the first leg focused mainly on interaction-based recommender systems, such as association rules, collaborative filtering, matrix factorization, BPR, and neural collaborative filtering, this second part explores whether semantic retrieval from movie metadata can yield useful, competitive recommendations.

The goal is to build a vector database from the metadata in the ml-latest-small dataset and use it to recommend movies based on semantic similarity. More specifically, given a current user session containing

different movies, the system should use the API access to IAEdu (or another API-based language model), to summarize the user's preferences. This summary will then be used as a query to retrieve semantically similar movies from the vector database.

This second leg should be understood as a content-based and IR-oriented extension of the first leg. In this case, the recommendation process will exploit textual information such as movie titles, genres, tags, and possibly additional metadata. The final objective is to compare the recommendations produced by the RAG-based pipeline with the ground truth and the best recommender systems from the first leg, as well as to provide some automatic explanations for the recommendations made.

This part also connects directly with the first assignment's requirement to follow a clear data science narrative: explain what is being done, why it is useful, how it is evaluated, and what conclusions can be drawn from the results

### Tasks (2nd leg)

- Build an interface to IAEdu, or another API-based language model, to summarize a current user session into a short query that captures the user's preferences. Test different prompts and consider the model's context window. If the full session is too large, use strategies such as selecting a representative sample of movies.
- Build a vector database using textual metadata from ml-latest-small, such as movie titles, genres, and tags. If useful, use links.csv to obtain additional metadata.
- Add a retriever to select recommendation candidates from the vector database, and optionally add a re-ranker to improve the final ranking. Compare different configurations, such as retriever only and retriever followed by re-ranking.
- Compare the RAG-based recommendations with the ground truth and with the best recommender systems developed in the first leg.
- Develop a module that uses an LLM to explain the recommendations from the second leg, based on the current user session. Evaluate a small sample of explanations using predefined criteria, such as usefulness, faithfulness, clarity, and specificity. Likert scales and predefined evaluation questions may be used.

## To submit:

- a fully operational Jupyter notebook with the selected experiments as clear and concise as possible. Avoid output dumps. Recall that the report is going to be evaluated by your very busy professors and that they may have to skip many pages if your report is too long. Always highlight your best results. Please note:
  - The objectives for each experiment and plot should be clear so that the reader understands why it is worth to read a particular part.
  - The conclusion should be a short high level account of what was observed.
  - It is **not necessary to describe the methods** (unless requested, but you should know their concepts and how they work). It is more important to point out the differences in the methods and the reasons for the results in terms of methods characteristics.
  - Describe succintly how you used external resources, including Generative AI.
- The project slides presentation.

## Evaluation

- This assignment is worth the values described in sigarra, according to the course you are following.
- Components
  - Report 30%

    - Narrative 10%
    - Writing style 10%
    - Presentation 10%
  - Technical 70%
    - Pre-processing 10%
    - Diversity of the results for the experiments 20%
    - Correctness 30%
    - Conclusions 10%
  - Individual presentation (**multiplied** by the sum of the two previous items)
    - Communication 50%
    - Technical proficiency (you know what you are talking about) 50%
- Member of the same group may have different grades. A group may also declare that different members have different effort rates.

## Groups

Assignments are submitted by groups of 3 students. Different elements may have different grades. Other group sizes will not be considered. If different elements of the group have different levels of effort that should be stated upfrint by the group in the report.

It is advisable that the students from the same group perform overlapping work and only after that, exchange ideas with each other. Group work is important for learning from other people.

## Submissions

Formal final deadline is **June 1st 2026**, to be submitted in moodle, and only in moodle. Submissions after that date will be multiplied by a monotonously decreasing factor that starts in 1 with a negative slope of .15 values per day until a minimum of 50% of the original value.

- Checkpoints:
  - In the class of the **27th April** there will be a checkpoint for this leg. Each group should present on desk an update with the status of the project. You will have around 5 minutes for this presentation.
  - Final presentations of the project on **June 5th** morning. All members of the group should be present. No-shows get a zero.

## Use of resources including Generative AI (pedagogical advise)

You, students, are learning how to do data science and the main aim of this exercise is that you gain skills. Skills are gained with different depth and robustness depending on how we use the learning resources. Effort must always be there. Beware of too easy routes - they usually teach us nothing. No pain, no gain.

Your **resources** are class materials, books, youtube videos, materials on the web, etc. (CM), programming documentation, such as the manuals of packages you use, search engines, e.g. google (SE) and generative AI, e.g. claude, mistral, chatgpt or gemini (GAI).

Our **advise** is that you approach your problems by using these resources by this order and with different types of usage. When thinking of concepts, use CM, when you need to know how to use a function or what to use in a particular setting, use PD. SE are very important for more directed searches, looking for code snippets, looking for more materials, finding fruitful discussions. Finally GAI gives us more value when used for concept clarification or to find the starting point for a search or quest. The more you know when you use GAI, the more you take from it. If you take nothing, you learn nothing. Many other usages of GAI are productive but they should always have you leading, not the other way around.

An **example of good usage of GAI** is "Which hyperparameters are important for using BPR and why?". A **bad usage** is "produce the code for applying BPR for this problem". You can gain skills with the first type of usage whereas with the second you rely entirely on the skills of generative AI. That implies that any future employee will rather buy a license of Claude or chatGPT than hire you (because it is cheaper). Or they will hire someone with more skills that can validate and better exploit GAI. That is because when you gain skills you are ready to gain more skills and you are a valuable asset that can check what gen AI is producing. Having skills makes you think longer, deeper and depend less on externalities. Deep skills make you more robust and more resilient. You will be valuable even without GenAI. If your skills are easy to get then anyone cheaper than you can do it.

## Ethical principles

When submitting, students commit themselves to follow strong ethical principles. All the work must be done by the elements of the group alone. All members of the group will be involved with the whole of the work. All the materials used and consulted must be credited in the work.